Ask Professor G. How to Grasp: Training-Free Cross-Gripper Grasping from an RGB-D Observation and a Gripper URDF

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***Abstract*— Grasping systems are increasingly expected to operate with heterogeneous end-effectors, from parallel-jaw grippers to multi-finger dexterous hands. While learning-based predictors remain tightly coupled to fixed embodiments and require costly retraining for novel URDF-specified grippers, optimization-based planners generalize across embodiments but struggle with non-convex contact dynamics and handcrafted objectives that fail to encode semantic intent.**

We present a training-free cross-gripper grasp synthesis framework driven by test-time optimization. An off-the-shelf foundation model, *Prof. G.*, provides semantic conditioning by generating structured search priors and executable objective specifications under a fixed domain-specific API. Instance-level refinement is performed via the Cross-Entropy Method over a non-differentiable energy landscape, enabling direct adaptation to novel URDF grippers without retraining. Experiments in simulation and on real hardware demonstrate improved grasp success and semantic compliance compared to representative geometry- and learning-based baselines. Code is available at `https://github.com/sunyuan1111/ASK-Professor-G-How-to-Grasp.git`.

I. INTRODUCTION

Robotic grasping has advanced rapidly with large-scale 6-DoF benchmarks and learned grasp perception systems that predict geometry-feasible grasps from RGB-D or point clouds [1], [2], [3], [4], [5]. However, a key bottleneck in practical manipulation is the diversity of end-effectors: a grasping system is expected to support heterogeneous grippers and hands, from parallel-jaw grippers to multi-finger dexterous hands. When a novel gripper is introduced via a new URDF, many learned grasp predictors degrade and typically require gripper-specific data collection, retraining, or nontrivial adaptation, incurring recurring retraining cost and deployment friction [6]. Although recent multi-gripper datasets expand hardware coverage [7], [8], gripper adaptation remains fundamentally training-driven and thus scales poorly with the long tail of end-effector designs.

At the same time, task execution increasingly demands semantic controllability rather than purely geometric stability: functional grasping often requires grasping specific parts and avoiding fragile or restricted regions. Learning-based language-conditioned grasping has made progress in grounding grasp proposals in semantics, but typically still relies on training with gripper-specific representations or supervision [7], [9]. Conversely, optimization-based grasp planning generalizes naturally across embodiments by explicitly searching for configurations that satisfy physical feasibility and analytic stability criteria [10], [11], but its objectives are usually hand-crafted and remain largely semantic-agnostic. This creates a central tension: *training-driven semantic grasping* does not remove cross-gripper deployment cost, while *embodiment-general optimization* does not natively encode semantic intent.



Fig. 1. Overview of the proposed training-free cross-gripper grasp synthesis framework. Given a single RGB-D observation and a URDF-specified gripper, the foundation model provides semantic conditioning to construct a strategy-conditioned search region and an executable objective, followed by CEM-based refinement for final grasp synthesis.

This work explores a complementary direction that resolves the above tension: *training-free cross-gripper grasp synthesis* via test-time optimization, where off-the-shelf foundation models are used for semantic conditioning and objective specification rather than direct grasp prediction. As illustrated in Fig. 1, the core insight is that grasp synthesis can be decomposed into two ingredients required at deployment: (i) a *search region* that captures plausible grasp strategies under a given gripper morphology, and (ii) an *objective* that evaluates candidates for both physical validity and semantic intent. In our implementation, these ingredients are produced by *Prof. G.*, a Gemini-family

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model accessed via commercial APIs. It outputs strategy-conditioned priors over a unified pose plus compact gripper-configuration parameterization [12], [9] and compiles semantic constraints into an executable energy function. To ensure reproducibility, objective evaluation is restricted to a fixed domain-specific API with a bounded set of primitives [13], [14]. Instance-level refinement is performed using the Cross-Entropy Method (CEM), enabling constrained optimization over non-differentiable energy landscapes across heterogeneous grippers.

Experiments in *MuJoCo* show clear gains over prior geometry- and learning-based baselines. In simulation, our method achieves a *success rate* (SR) of 21.28% on WSG-50 and 63.68% on Allegro, improving SR by up to 11%. For semantic tasks, it maintains a strong *semantic success rate* (SSR), indicating that successful grasps consistently satisfy the specified semantic constraints.

In summary, the contributions of this work are threefold:

- A training-free formulation of cross-gripper grasp synthesis that enables direct deployment on novel URDF-specified end-effectors without gripper-specific retraining.
- A semantics-driven conditioning mechanism that converts heterogeneous gripper morphology and a single RGB-D observation into strategy-conditioned priors over a unified pose–configuration grasp space.
- An API-constrained objective compilation and CEM-based refinement framework that translates semantic intent into a reproducible, executable energy function across heterogeneous grippers.

II. RELATED WORK

A. Learning-based Grasp Perception

Learning-based grasp perception has progressed rapidly with large-scale 6-DoF benchmarks and increasingly expressive models. Early data-driven approaches predict geometry-feasible grasps from RGB-D or point clouds, typically assuming a fixed end-effector. Representative systems such as GraspNet-1Billion [1] and Contact-GraspNet [2] learn dense SE(3) grasp proposals for cluttered scenes, while later methods improve robustness and efficiency but largely remain focused on parallel-jaw grippers, including AnyGrasp [3] and equivariant designs such as Edge Grasp Network [15].

Despite these advances, generalization across heterogeneous grippers remains a fundamental limitation. Most learned predictors implicitly couple perception, collision priors, and grasp scoring to a specific kinematic structure. When deployed on a novel URDF-specified gripper, performance typically degrades and requires either gripper-specific retraining [6] or additional conditioning modules that still relearn embodiment-dependent feasibility [7]. Although recent efforts introduce multi-gripper datasets and gripper-aware generation mechanisms [7], [8], [9], these approaches remain training-driven and constrained by dataset coverage.

Language- and foundation-model-conditioned grasping further introduces semantic grounding beyond geometry. Methods such as Grasp-Anything [16], LGD [17], mask-guided attention [18], and GraspMamba [19] incorporate language or segmentation cues, while more recent systems leverage reasoning-based vision–language models, including ORACLE-Grasp [20], AffordGrasp [21], and VCoT-Grasp [22]. However, these methods generally remain training-dependent and tied to fixed or annotated embodiments, making adaptation to novel heterogeneous URDF grippers costly at deployment.

These limitations motivate training-free cross-gripper grasp synthesis that integrates semantic constraints without recurring retraining when the end-effector changes.

B. Optimization-based Grasp Planning

Optimization-based grasp synthesis searches for $\mathbf{g}^*$ that minimizes an objective $J(\mathbf{g}, \mathcal{O})$ under object representation $\mathcal{O}$, leveraging explicit geometric and physical criteria and thus generalizing well across embodiments. Classical metrics such as force closure and the ϵ-quality measure [10] motivate stability evaluation in wrench space, and simulators such as GraspIt! [11] perform heuristic search for stable configurations. To cope with high-dimensional search spaces, sampling-based refinement methods, including CEM and Metropolis-Hastings, are widely used [23].

Recent work advances toward optimization with richer representations and stronger priors. Differentiable formulations enable gradient-based refinement, including bilevel objectives that couple hand kinematics and contact forces for high-DoF hands [24]. To relax the need for precise meshes, some methods optimize directly on point clouds or implicit surfaces; for example, GrainGrasp [25] exploits contact cues extracted from point cloud geometry. Trajectory-level methods further integrate grasp stability into motion planning [26]. Despite these advances, most optimization-based planners remain largely semantics-agnostic and do not natively encode functional constraints.

C. Foundation Models for Reward Generation

LLMs have recently been used to synthesize executable reward functions or policies, shifting from end-to-end learning to programmatic specification. Code-as-Policies (CaP) [13] composes perception APIs and motion primitives for long-horizon tasks, building on earlier high-level skill sequencing such as SayCan [27]. Subsequent work extends LLM-based reasoning into continuous spatial domains, e.g., VoxPoser [14] for 3D value-map construction. However, these methods largely target task-level planning or assume fixed embodiments, and do not address training-free, low-level grasp synthesis for heterogeneous URDF-specified grippers.

III. METHOD

Throughout this paper, *LLM* and *VLM* refer to off-the-shelf foundation models accessed via commercial APIs with fixed parameters. In our implementation, *Prof. G.* is instantiated using `gemini-3.0-pro-preview` (API version at the time of experiments), which supports multimodal image–text

inputs. We used this backend for stable multimodal conditioning and structured JSON/Python outputs; the interface can use any model that returns the same schema and `GraspLib`-only code, but all reported quantitative results use the Gemini-family backend. The model is employed solely for semantic conditioning and objective compilation, i.e., generating a structured *Search Region* and an executable *Objective Function* (Fig. 2). All adaptation to a novel URDF-specified gripper and the observed scene is achieved through instance-level optimization over grasp variables, rather than learned prediction. At inference time, both the foundation model and the compiled objective are conditioned only on the RGB-D observation and the gripper URDF; no ground-truth part labels, masks, keypoints, or annotation-derived semantic regions are provided or accessed during grasp generation or CEM optimization.

Example trace. In a lamp–WSG50 case, *Prof. G.* outputs three visual proposals, expands them into ten pose–synergy boxes, compiles target/contact/collision terms into `calculate_loss`, and CEM selects the lowest-loss grasp.

A. Semantics-Driven Grasp Search Region Initialization

Object Information and Object Semantics. Given a single RGB-D observation, we construct *Object Information* as (*Object Image*, *Calibration Data*) to condition the foundation model (Fig. 1). The *Object Image* is the RGB image, augmented with a lightweight axis overlay to make the camera frame explicit. The accompanying *Calibration Data* provides the metric grounding required to lift image-space predictions into 3D: we use calibrated intrinsics to back-project depth into a partial point set and derive coarse geometric cues (e.g., the visible-region centroid and the 3D bounding-box size), and when a world/base frame is needed for execution, we additionally apply the extrinsic transform $\mathbf{T}_{\text{world}\leftarrow\text{cam}}$. Conditioned on this object-level input, the model predicts *Object Semantics* that are directly actionable for grasping such as functional target/forbidden regions and affordance cues, which in turn specify a *Search Region* rather than a single grasp. Concretely, the output semantics define where grasping should (and should not) occur on the object, thereby constraining the downstream sampling of *Grasp Poses* and *Synergy Values* within the CEM optimizer.

Gripper Information and Search Region over Pose–Synergy Space. To generalize across heterogeneous end-effectors without gripper-specific learning, we provide *Gripper Information* extracted from the URDF through the modules in Fig. 2. The *Gripper Extractor* converts the URDF into a compact, semantically explicit *Gripper Config* describing the kinematic topology, functional axes (e.g., palm normal $\mathbf{n}_{palm}$ and closing direction), and physical limits such as maximum aperture that rule out infeasible contacts. In parallel, the *Synergy Extractor* maps the original high-dimensional configuration into low-dimensional *Synergy Variables* using Principal Component Analysis (PCA) [12], [9]. We compute PCA synergies from a small set of canonical closing configurations generated solely by sweeping a scalar closure command within URDF joint limits (no grasp demonstrations). We then parameterize each grasp candidate by a unified pose–synergy state $\mathbf{g} = [\mathbf{p}, \boldsymbol{\theta}, \mathbf{s}]^\top$, where $\mathbf{p} \in \mathbb{R}^3$ and $\boldsymbol{\theta} \in \mathbb{R}^3$ represent the end-effector translation and Euler-angle orientation, and $\mathbf{s} \in \mathbb{R}^d$ denotes the synergy coefficients (*Synergy Values*). Conditioned on *Object Information* and *Gripper Information*, the foundation model produces *Search Region* as priors over *Grasp Poses* and *Synergy Values*. In practice, the model first localizes coarse semantic anchors in the image, which are back-projected into initial 3D positions. The remaining degrees of freedom are initialized by a hybrid rule that reduces geometric drift: the approach direction is constrained by a local surface normal $\mathbf{n}_{surf}$ estimated from depth, enforcing $\mathbf{n}_{palm} \approx -\mathbf{n}_{surf}$, while the model infers only the semantically sensitive components such as the roll angle and the synergy coefficients $\mathbf{s}$. The resulting distribution over $\mathbf{g}$ provides a robust warm start for the subsequent CEM-based instance-level refinement.

B. LLM-Synthesized Objective Compilation

Conventional grasp planners typically optimize fixed, manually designed energy functions, which makes it difficult to incorporate task-specific semantic requirements, such as avoiding fragile components or favoring particular affordance regions. To address this limitation, we use an LLM as a *semantic compiler* that converts perceptual observations and task context into an executable objective $J(\mathbf{g})$ for guiding downstream black-box optimization.

To guarantee deterministic execution and reproducible evaluation, we constrain code synthesis to a fixed, domain-specific API boundary. Specifically, the LLM is restricted to a sandboxed interface, denoted as `GraspLib`. This interface exposes a finite set of atomic primitives organized into four functional modules: (i) spatial transformations for coordinate conversion; (ii) geometric analysis for local shape and distance computation; (iii) feasibility checking for collision and proximity evaluation; and (iv) data utilities for point-cloud preprocessing. In our implementation, `GraspLib` exposes 16 bounded primitives, including coordinate transforms, point filtering, alignment scoring, and collision checks. This constraint prevents arbitrary operations and abstracts numerical implementation details, enabling the LLM to synthesize high-level task logic into deterministic computations.

To minimize computational overhead during sampling-based optimization, we implement a pre-compilation protocol. Rather than iteratively parsing perceptual context at runtime, the LLM extracts task-relevant parameters once and embeds them as global constants within the generated code. The compiled objective scores a candidate grasp state $\mathbf{g}$ via the generated function:

$$J(\mathbf{g}; \mathcal{G}) \;=\; \text{Objective}(\mathbf{g}, \mathcal{G}), \tag{1}$$

where $\mathcal{G}$ denotes the point-based geometric representation processed by `GraspLib`. Internally, $J(\mathbf{g}; \mathcal{G})$ typically comprises two types of terms: geometric and semantic. The geometric terms aggregate physical feasibility and grasp-quality criteria, including collision/interpenetration penalties

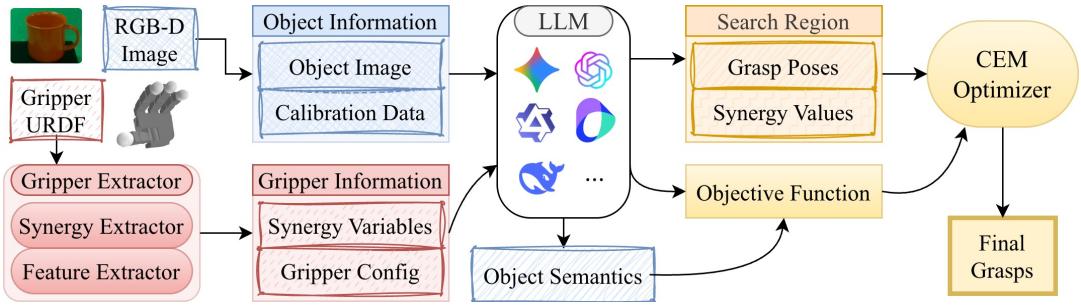


Fig. 2. System architecture for our grasp synthesis. The diagram illustrates the dual object–gripper information streams, LLM-based compilation of search regions and objective functions, and subsequent CEM optimization for final grasp generation.

and stability-related measures, while the semantic terms encode the compiled preferences such as target-zone attraction and forbidden-zone avoidance.

C. Instance-Level Refinement via CEM

While the semantic prior (Sec. III-A) delineates the search region and the synthesized objective (Sec. III-B) defines the cost landscape, identifying the precise optimal configuration requires numerical refinement. We formulate grasp synthesis as an energy minimization problem over the joint grasp state $\mathbf{x} = [\mathbf{p}, \boldsymbol{\theta}, \mathbf{s}]$, where $\mathbf{p}$, $\boldsymbol{\theta}$, and $\mathbf{s}$ denote the wrist translation, Euler-angle orientation, and PCA synergy coefficients, respectively.

The compiled objective function combines discrete logical checks, such as collision pruning, with continuous geometric primitives from `GraspLib`. As a consequence, the resulting energy landscape is non-convex, non-differentiable, and potentially discontinuous. Consequently, gradient-based methods are ill-suited for this task. We therefore employ the Cross-Entropy Method (CEM), a robust zeroth-order importance sampling algorithm capable of optimizing black-box functions under complex constraints.

For each grasp strategy proposed by *Prof. G.*, we run an independent CEM refinement and treat the strategy-conditioned priors as a multi-modal initialization. Each run maintains a factorized Gaussian distribution over the grasp variables, initialized from the model-predicted bounds, and iteratively samples candidates, evaluates them with the compiled objective, and updates the distribution using the elite set. Hard box constraints are enforced by clipping samples to the valid ranges. After a fixed number of iterations, we select the minimum-objective solution across all strategies as the final grasp.

IV. EXPERIMENT

A. Experimental Setup

Tasks, environments, and metrics. We evaluate our method in simulation and on real hardware. In simulation, we measure grasp success rate, contact and stability metrics, and semantic compliance with target- and forbidden-zone constraints. On hardware, we validate end-to-end lift-and-place execution, considering a trial successful if the object is grasped and placed at the designated location, and we also assess semantic compliance under the same task specification.

We report the following metrics throughout the experiments, each computed per object (aggregated over repeated trials) and then averaged across all objects. *Success Rate* (SR, %) is the fraction of trials that satisfy the stable holding or lift-and-hold success criterion. *Contact Distance* (CD, mm) measures the mean contact gap after closure, computed as the average distance between the gripper contact elements and the object surface at the final grasp configuration (smaller indicates tighter contact). *Penetration Distance* (PD, mm) is the maximum interpenetration depth observed during simulation execution, used as a feasibility indicator. *Contact Count* (CC, count) denotes the number of valid contacts established between the gripper and the object. *Grasp Wrench Space* (GWS, unitless) is a scalar stability proxy derived from the grasp wrench space constructed from the observed contacts, using a fixed Coulomb friction coefficient $\mu = 1$. *Repeat Grasp Rate* (RGR, %) quantifies grasp diversity as the fraction of near-duplicate grasps under a fixed similarity threshold in the joint pose–synergy space. For semantic tasks, we report *Target Zone Compliance* (TZ, %) and *Forbidden Zone Avoidance* (FZ, %), as well as the *Semantic Success Rate* (SSR, %) which counts successful grasps that also satisfy all semantic constraints. For semantic metrics (TZ/FZ/SSR), target/forbidden zones are defined only for offline scoring using part-level annotations when available; these annotations are never provided to the method during grasp generation or CEM optimization.

Object set. Our object set contains 108 models drawn from three sources: 65 objects from MultiFingerGrasp [7], 33 objects from DexGraspNet [8], and 10 custom-designed assets. For a subset of objects, we additionally have part-level semantic annotations (e.g., handle or rim regions) for functional grasping. The simulation semantic benchmark uses nine annotated object instances spanning containers, cookware, teapots, and tools. These annotations are used only for evaluation of semantic compliance (e.g., TZ/FZ/SSR) and are never provided to *Prof. G.* at inference time. The remaining objects do not provide semantic annotations and are treated as geometry-only instances (e.g., plush toys). In real-world experiments, we select only objects for which such annotations are available, comprising seven different

cups and five different tools, as illustrated in Fig. 5.

Gripper set and categories. We evaluate seven end-effectors spanning three categories to test cross-morphology generalization: (i) two-finger parallel-jaw grippers, (ii) three-finger grippers, and (iii) multi-finger dexterous hands. The evaluated set covers six commercially available grippers/hands (Allegro [28], Barrett [29], FreedomHand [30], Kinova Jaco [31], Robotiq 3F Adaptive Gripper [32], and Weiss WSG-50 [33]), together with one human hand model [7].

Optimization settings. We use CEM with N_{cem}=500 samples/iter, E_{cem}=50 elites, K_{cem}=20 iterations, EMA smoothing α=0.7, and a per-dimension std floor $\sigma_{\min}=10^{-4}$ (with box-clipping to valid ranges).

Baselines. Baselines are grouped into geometry-based and learning-based methods. *Geometry-based* baselines include: (i) **Normal-Sampling (NS)**, which samples parallel-jaw grasp candidates by aligning the closing direction with local surface normals and filters candidates with geometric feasibility checks; and (ii) **Grasp Pose Generator (GPG)** [34], [35], a grasp pose detection pipeline that generates and ranks parallel-jaw grasp hypotheses from point clouds in clutter. *Learning-based* baselines include **UniDexGrasp (UDG)** [36] and **DexDiffuser (DD)** [37], which generate dexterous grasp proposals from point-cloud observations using learned proposal generation mechanisms.

B. Results

We compare our method with geometry-based and learning-based baselines in *MuJoCo*. Qualitative results in simulation are shown in Fig. 4. The top panel presents semantic-aware goal-directed grasps guided by model-predicted target and forbidden zone constraints, while the bottom panel shows grasps generated without semantic guidance.

The optimization dynamics are illustrated in Fig. 3. The left panel presents the distribution of the best objective values at different stages of the optimization process, including the initial sampling, the final CEM-optimized results, and the dropped candidates. A clear shift toward lower objective values can be observed after optimization, indicating that CEM effectively refines the grasp configuration under the compiled objective. The right panel shows the iteration-wise convergence curves for 10 randomly selected grasps. In most cases, the objective value decreases steadily within the first few iterations and gradually stabilizes, demonstrating consistent convergence behavior of the optimization procedure.

NS and GPG [34] are designed specifically for two-finger parallel-jaw grippers and are therefore evaluated only with WSG-50. We train and evaluate UDG [36] and DD [37] separately on two representative grippers, Allegro and WSG-50. The two models are trained on the Mgg dataset [7] following the official train/test split. Under the unified protocol in Sec. IV-A, our method achieves the highest grasp success rate on both grippers under this protocol. We note that this comes with a trade-off: since the compiled objective prioritizes stable lift-and-hold success and compliance with

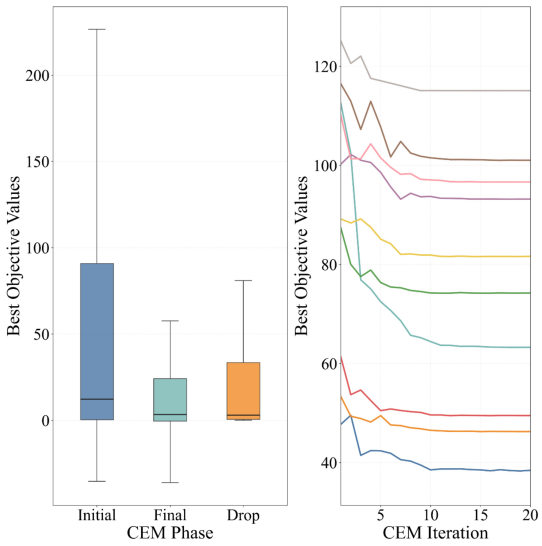


Fig. 3. Optimization behavior in simulation. Left: Boxplot of best objective values at different CEM stages. Right: Convergence curves of CEM over 20 iterations for 10 randomly selected grasps.

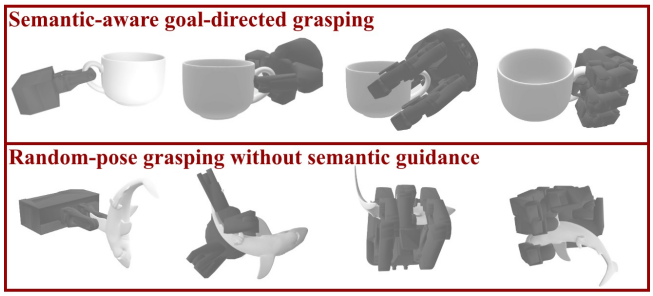


Fig. 4. Qualitative results of synthesized grasps in *MuJoCo* simulation.

target/forbidden semantic constraints, the optimized grasps may exhibit larger geometric diagnostic values compared to baselines that explicitly regularize tight surface contact or minimize interpenetration. UDG [36] performs poorly on WSG-50 mainly because it is specifically designed for dexterous multi-finger hands and does not transfer well to two-finger parallel-jaw grippers. All results are reported using the same set of metrics in Sec. IV-A.

Table II reports per-gripper performance of our method, where each metric is computed per object and then averaged across objects for the corresponding gripper. Overall, the method remains robust across heterogeneous morphologies, with dexterous hands typically achieving higher contact richness and stability, while parallel-jaw grippers exhibit slightly lower CC and GWS on geometrically complex objects.

Table III evaluates semantic grasping under specified target and forbidden zones, reporting TZ, FZ, SR, and SSR. Across grippers, FZ exceeds 90% in most cases, while TZ varies across grippers and is typically in the 20–70% range for encouraged grasp regions. To ensure consistency, all metrics in this table are computed exclusively on the subset of objects with semantic annotations.

TABLE I

PERFORMANCE AND BASELINE COMPARISON IN *MuJoCo*.

Method	SR ↑	CD ↓	PD ↓	CC ↑	GWS ↑	RGR ↓
WSG-50						
NS	9.37%	5.9	9.5	0.91	0.0010	1.32%
GPG [34]	10.23%	6.3	8.9	0.85	0.0008	14.38%
UDG [36]	3.08%	8.3	8.6	0.55	0.0003	**0.00%**
DD(pn2) [37]	10.91%	3.6	4.5	1.19	0.0017	3.12%
DD(bps) [37]	11.09%	**3.5**	**4.5**	1.20	0.0016	3.08%
Ours	**21.28%**	7.3	7.7	**1.43**	**0.0020**	8.51%
Allegro						
UDG [36]	24.62%	6.5	7.5	2.56	0.0161	**0.00%**
DD(pn2) [37]	30.91%	4.7	6.2	2.81	0.0229	1.00%
DD(bps) [37]	49.36%	**4.4**	**6.0**	4.03	0.0370	1.40%
Ours	**63.68%**	7.9	8.4	**5.24**	**0.0426**	3.78%

TABLE II

CROSS-GRIPPER EVALUATION IN *MuJoCo*.

Gripper/Hand	SR ↑	CD ↓	PD ↓	CC ↑	GWS ↑	RGR ↓
Parallel-jaw grippers						
WSG-50	21.28%	7.3	7.7	1.43	0.002	8.51%
Three-finger grippers						
Barrett	26.96%	7.4	8.0	2.74	0.0218	5.65%
Robotiq-3F	35.68%	9.4	9.9	3.98	0.0464	4.15%
Jaco	67.98%	7.6	8.2	4.67	0.0302	5.26%
Multi-finger dexterous hands						
Allegro	63.68%	7.9	8.4	5.24	0.0426	3.78%
FreedomHand	47.70%	7.0	7.4	6.10	0.0384	6.90%
HumanHand	25.32%	7.2	8.0	3.04	0.0192	2.58%

TABLE III

CROSS-GRIPPER GRASP SUCCESS AND SEMANTIC CORRECTNESS.

Gripper	TZ ↑	FZ ↑	SR ↑	SSR ↑
Parallel-jaw grippers				
WSG-50	30.00%	93.33%	16.67%	13.33%
Three-finger grippers				
Barrett	24.14%	89.66%	20.69%	13.79%
Robotiq-3F	44.44%	88.89%	27.78%	22.22%
Jaco	72.41%	96.55%	65.52%	58.62%
Multi-finger dexterous hands				
Allegro	26.47%	97.06%	52.35%	40.59%
FreedomHand	42.86%	95.24%	42.86%	33.33%
HumanHand	21.21%	100.00%	18.18%	15.15%

TABLE IV

ABLATION STUDY.

Method	SR ↑	CD ↓	PD ↓	CC ↑	GWS ↑	RGR ↓
w/o Axis Cue	38.13%	7.4	8.0	3.67	0.0264	5.22%
w/o Step 2	36.89%	**6.6**	**7.2**	3.27	0.0229	7.98%
w/o Step 2/3	28.31%	7.0	7.7	2.99	0.0186	**2.74%**
Full method	**41.23%**	7.7	8.2	**3.89**	**0.0286**	5.26%

C. Ablation Study

We evaluate the contribution of each stage in our pipeline: Search Region Initialization (Step 1), Objective Compilation (Step 2), and CEM Refinement (Step 3). Table IV reports three ablation variants. (i) *w/o Axis Cue* removes only the Step 1 visual axis overlay, so it is a localization-aid ablation rather than removal of the full initializer. (ii) *w/o Step 2* replaces the LLM-compiled objective with a fixed hand-crafted objective while retaining the optimization procedure. (iii) *w/o Step 2/3* removes both objective compilation and optimization, directly executing the output from Step 1. The *Full method* corresponds to the complete pipeline. The full method achieves the highest SR, CC, and GWS. Removing the axis overlay only slightly reduces SR and grasp quality, indicating that this visual cue mainly provides auxiliary spatial guidance rather than being the dominant performance factor. Replacing Step 2 with a fixed objective leads to a noticeable drop in SR and GWS due to reduced objective flexibility and limited adaptation to instance-specific geometry. Interestingly, CD and PD slightly improve under this setting, suggesting that a fixed objective enforces stronger geometric regularization and marginally enhances physical plausibility, albeit at the expense of grasp success. Finally, removing both Step 2 and Step 3 results in a substantial degradation across success and contact-related metrics, confirming that test-time objective compilation and optimization are essential for accurate and physically consistent grasp synthesis.

D. Real World Experiments

We conduct real-world experiments to validate the system's performance on physical hardware. The experiments involve three distinct setups: (i) a Universal Robots UR5 [38] arm equipped with a FreedomHand [30], (ii) a UR5 arm with a DaHuan AG95 [39] gripper, and (iii) the SOARM101 robot system [40]. In all setups, perception is provided by Intel RealSense D435 cameras [41]. Setups (i) and (ii) employ a hybrid configuration with one eye-in-hand camera and one external eye-to-hand camera, while Setup (iii) uses two external eye-to-hand cameras due to the limited payload capacity of SOARM101. The experimental setups and the object set are illustrated in Fig. 5.

For each setup, we evaluate performance across different object groups, attempting 3 trials per object configuration. We report the average SR and SSR for each group. As shown in Table V, all setups consistently achieve SR above 88%, with several groups reaching 100% success. SSR closely follows SR, remaining above 78% across all setups and exceeding 90% in most tool categories. Qualitative results of real-world grasps are shown in Fig. 6.

The real-world evaluation uses a smaller, curated set of readily graspable objects and executes only the single best grasp per trial, whereas simulation evaluates multiple strategy-conditioned candidates and reports aggregate metrics over a broader object distribution. These differences in object diversity and evaluation protocol mean that the

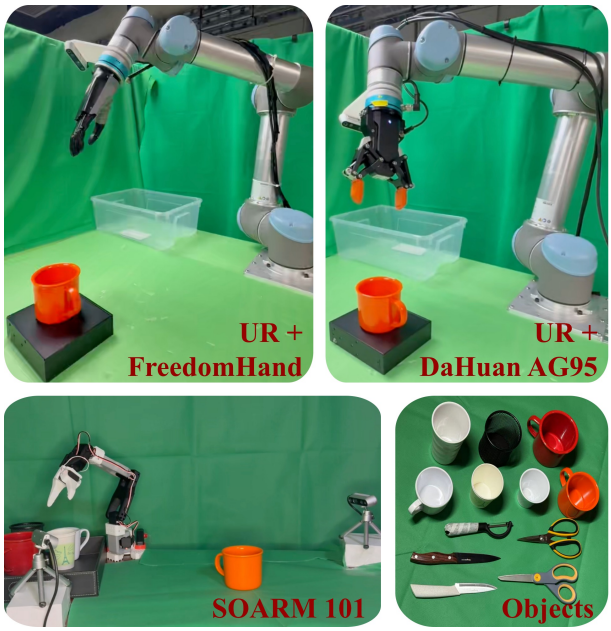


Fig. 5. Real-world experimental setups with three configurations and the object set.

TABLE V
REAL WORLD EXPERIMENTS.

Setup	Group	SR ↑	SSR ↑
UR+FreedomHand	Tool 1	88.24%	88.24%
UR+FreedomHand	Tool 2	92.31%	92.31%
UR+FreedomHand	Cup	90.00%	81.82%
UR+DaHuan AG95	Tool 1	93.75%	93.75%
UR+DaHuan AG95	Tool 2	100.00%	100.00%
UR+DaHuan AG95	Cup	100.00%	86.67%
SOARM101	Tool	93.75%	93.75%
SOARM101	Cup	88.24%	78.33%

absolute SR/SSR numbers in simulation and in the real world are not directly comparable.

V. CONCLUSIONS

This paper presents a training-free framework for cross-gripper grasp synthesis that supports heterogeneous URDF-specified end-effectors without retraining. By combining foundation-model-based semantic guidance with test-time optimization, the method produces grasps that satisfy both physical constraints and task-level requirements across different embodiments. Experimental results in simulation and on real hardware demonstrate improved grasp success and semantic compliance over representative baselines.

Limitations include the latency and potential variability of commercial foundation-model APIs, as well as reliance on part-level semantic specifications for evaluating semantic compliance. Future work will explore more efficient prompting and caching strategies, stronger safeguards for objective synthesis under API/model updates, and broader semantic grounding that reduces dependence on explicit part annotations while extending validation to more diverse sensing conditions and hardware platforms.

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Fig. 6. Qualitative results of real-world grasps.

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